



Verify Application Users in Two Simple Steps

This article explains the two-step verification process, and how Java application developers can implement it in their apps, using NetBeans and JavaMail API. The author assumes that readers are familiar with core Java and Java Swing, and have a working knowledge of NetBeans.

We live in a time when the Internet has become a necessity as basic as food, shelter or even the air we breathe. While in India this may not be true for 60 per cent of the rural population, those who live in the urban world would certainly agree. Given the amount of time we spend online, it only makes sense to increase the levels of security and privacy of our data and activities on the Internet. Not more than two or three years back, gaining access to someone's email and social network accounts was just a matter of basic social engineering. But with time, service providers like Google, Facebook, etc, realised their weak spots, and introduced various new techniques that make sure that our data remains ours.

Many readers, who are also avid Internet users, must be familiar with the two-step verification process. While the concept has been in use for quite some time, earlier its usage was limited to things that were considered to be highly confidential—like government facilities, military set-ups or data centres. In this article, I'll demonstrate how this feature can be used in your Java applications. Before moving further, let's understand the two-step verification with the help of a story.

What is two-step verification?

Imagine you have a friend with a big mansion that has a roof-top swimming pool, a big lawn, 12 bedrooms, etc. Don't forget that security guard, who is provided with an intercom and a gun, and his cabin at the main gate. One fine day, you decide to pay a visit to this friend and show up at his main gate, where the security guard stops you and asks for your identity. You tell him your name and your relationship with the owner of the mansion (in this case, your friend). He calls up your friend on the intercom and informs him about your arrival. Your friend tells him to let you in and the gates open for you.

What just happened? You passed the first step of verification. But are you inside the mansion yet? No. So how will you get inside? Once inside the gates, you get to the door and ring the bell. Your friend hears the bell, comes down to the door, and after confirming that it really is you, opens the door and lets you in.

What happened here was that your friend saw your face and recognised you. What was the need for that when you already identified yourself at the gate? Recall that at the